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Task Description:	The neural crest and the cranial ectodermal placodes give rise to many of the defining characters of vertebrates, including a well-defined head with teeth and paired sensory organs. Both are migratory, form at the border between neural plate and epidermis, and generate some identical derivatives such as sensory neurons. To explore the events that led to the evolution of vertebrate features, this research will examine: (1) the evolutionary origin of the neural crest and placodes and (2) the interrelationship and evolutionary divergence of these two important embryonic structures. We aim to look for evolutionary precursors to neural crest and placodes in the invertebrate chordate, amphioxus, and to study neural crest and placodes in the primitive vertebrate, lamprey. We will utilize vital dye labeling to examine whether amphioxus has migratory cells that resemble neural crest/placodal cells in vertebrates. We will correlate the sources of migratory cells with gene expression domains of homologs to vertebrate molecular markers for neural crest cells and		

	placodes. In addition, we will map neural crest migratory pathways and derivatives in the lamprey. These represent questions of major importance in the evolution of the vertebrate body plan. Examine evolutionary origin of vertebrate neural crest and placodes, with emphasis on ear. Important for understanding vertebrate origins as well as evolution of vestibular system.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the origin of life on earth is critical to understanding if life may have evolved elsewhere in the universe. This research promises to be highly informative regarding the origin of vertebrate body plan. Neural crest cells and placodes are unique to vertebrates and are some of their most defining characteristics. However, little is known about how these tissue types evolved. By understanding their evolutionary origin from non-vertebrate chordates, we hope to better discern how the vertebrate body plan arose.
Task Progress:	1. Conservation and Divergence of BMP24 Genes in the Lamprey: Gene Expression Patterns Aid Phylogenetic Analysis Bone morphogenetic protein (BMP) molecules are members of a large family of signaling molecules important in numerous developmental pathways throughout the metazoa. Single members of the BMP2/4 class have been found in invertebrates such as cnidarians, arthropods, nematodes, echinoderms, ascidians, and cephalochordates. In all vertebrates studied, there are at least two copies, BMP2 and BMP4, which play important roles in axial patterning, tissue specification, and organogenesis. The basal vertebrate, lamprey, diverged near the time of vertebrate origins and is useful for understanding the gene duplication events that lead to the increased complexity of the vertebrate genome. We have characterized the sequence and expression pattern of BMP2/4 class genes in the sea lamprey, Petromyzon marinus. We uncovered three copies that we have named PmBMP24A, PmBMP24B, and PmBMP24C. Phylogenetic analysis indicates that PmBMP24A is closer than PmBMP24B or C in sequence identity to both BMP2 and BMP4 of gnathostomes. The developmental expression pattern of PmBMP24A also more closely resembles the combined early expression patterns of gnathostome BMP2 and BMP4, while PmBMP24B and C appear to play roles only later in development. Taken together, our sequence and expression data for PmBMP24A support the subfunctionalization of duplicated gnathostome BMP2 and BMP4 genes from a single BMP2/4 ancestral gene, and suggest that the duplication of BMP2/4 genes in lamprey occurred after the divergence of agnathans and gnathostomes. 2. Mandibular arch morphogenesis and the origin of jaws: new interpretation of “maxillary” and “mandibular” Cartilage of the vertebrate jaw is derived from cranial neural crest cells that migrate to the first pharyngeal arch and form a dorsal “maxillary” and a ventral “mandibular” condensation. It has been assumed that the former gives rise to palatoquadrate and the latter to Meckel’s (mandibular) cartilage. In anamniotes, these condensations have been thought to form the framework for the bones of the adult jaw and, in amniotes, appear to prefigure the maxillary and mandibular facial prominences. Here, we directly test the contributions of these neural crest condensations in axolotl and chick embryos, as representatives of anamniote and amniote vertebrate groups, using molecular and morphological markers in combination with vital dye labeling of late-migrating cranial neural crest cells. Surprisingly, we find that both palatoquadrate and Meckel’s cartilage derive solely from the ventral “mandibular” condensation. In contrast, the dorsal “maxillary” condensation contributes to trabecular cartilage of the neurocranium and forms a part of the frontonasal process but does not contribute to the jaw joint as previously assumed. These studies are the first to reveal the morphogenetic processes of cranial neural crest cells within the first arch that build the primordia of jaw cartilages and of the anterior cranium. 3. Id Gene Expression in Amphioxus and Lamprey Highlights the Role of Genetic Cooption During Neural Crest Evolution Neural crest cells are unique to vertebrates and generate many of the adult structures that differentiate them from their closest invertebrate relatives, the cephalochordates. Id genes are robust markers of neural crest cells at all stages of development. We compared Id gene expression in amphioxus and lamprey and asked if cephalochordates deploy Id genes at the neural plate border and dorsal neural tube in a manner similar to vertebrates. Furthermore, we examined whether Id expression in these cells is a basal vertebrate trait or a derived feature of gnathostomes. We found that while expression of Id genes in the mesoderm and endoderm is conserved between amphioxus and vertebrates, expression in the lateral neural plate border and dorsal neural tube is a vertebrate novelty. Furthermore, expression of lamprey Id implies that recruitment of Id genes to these cells occurred very early in the vertebrate lineage. Based on expression in amphioxus we postulate that Id cooption conferred sensory cell progenitor-like properties upon the lateral neurectoderm, and pharyngeal mesoderm-like properties upon cranial neural crest. Amphioxus Id expression also supports an evolutionary relationship between the anterior neurectoderm of amphioxus and the presumptive placodal ectoderm of vertebrates. We relate these observations to previous studies and propose that neural crest evolution was driven in large part by cooption of multi-purpose transcriptional regulators from other tissues and cell types. 4. Subcontract—Dr. Linda Holland The goal of our NASA-supported research has been to understand changes at the gene level that resulted in the evolution of vertebrate neural crest, which contributes essential elements to such important vertebrate features as the inner ear. Invertebrate chordates (tunicates and amphioxus) lack neural crest even though, like vertebrates, they have a dorsal, hollow nerve cord, which expresses many of the same genes in similar patterns as in vertebrates. To understand the innovations at the gene level that endowed cells at the edges of the neural plate with the ability to migrate away from the neural tube and to differentiate into such structures as the cartilage of the inner ear, we have focused on the FoxD genes in the forkhead family of transcription factors. During the transition from invertebrate to vertebrate chordates, large-scale gene duplications occurred resulting in 5 FoxD genes in vertebrates compared to only one in the invertebrate chordate amphioxus. Of these, only FoxD3 is expressed in premigratory neural crest cells at the edges of the neural plate as well as in migratory neural crest and has been shown to have a role initiation of neural crest cell migration and in differentiation of neural crest cells. Our work supported under this NASA grant showed that AmphiFoxD is not expressed in the edges of the neural plate in amphioxus (Yu et al., 2002). Therefore, we asked how FoxD genes have changed subsequent to gene duplication in order to allow new roles in neural crest. We have considered two possibilities. One is that new cis-regulatory elements directing expression to the edges of the neural plate evolved in the vertebrate FoxD3 gene subsequent to duplication of the FoxD genes in the vertebrates. The second is the FoxD3 protein changed as well, allowing the FoxD3 gene to perform new functions in cell migration and differentiation. For the first objective, we cloned 6.3 kb of upstream AmphiFoxD regulatory sequence into a lacZ reporter vector and expressed it in amphioxus.

	Methods for microinjection of amphioxus eggs that we developed are in press in “Methods in Cell Biology (Holland and Yu, 2004). This work, (Yu et al., 2004) is in press in Developmental Biology. In brief, we showed that this 6.3 kb region of upstream regulatory DNA is sufficient for directing expression to the three tissues that normally express the gene—the notochord, somites and anterior nerve cord. In addition, we identified a 4.7 kb region that was sufficient for directing expression to the notochord. In addition, we suggested in this paper that a cluster of binding sites for the notochord marker, brachyury might be essential for notochordal expression. Recently, we found that this is true. A construct with these sites deleted expresses in the somites, but not in the notochord. In addition, we suggested that the region with 2.5 repeats of a 268 bp motif might be an autoregulatory enhancer. Recently, we have found that it is indeed an autoregulatory enhancer. A construct with this region deleted does not express in amphioxus eggs. In collaboration with Dr. Mark Reedy (Creighton University), we have cloned the 6.3 kb construct into an expression vector suitable for the chick and electroporated it into the neural tube of chick embryos. As we did not see any expression with this construct, we conclude that cis-regulatory elements capable of directing expression to neural crest probably evolved in the FoxD3 gene subsequent to duplication of the vertebrate FoxD genes, although, of course, we cannot rule out the possibility that a region of the FoxD3 gene either upstream or downstream of the one we tested possesses such regulatory elements. These experiments are being continued in collaboration with Marianne Bronner-Fraser. To test the possibility that the function of the FoxD protein has also changed to allow a new role in promoting neural crest, in collaboration with Marianne Bronner-Fraser, we overexpressed AmphiFoxD in the neural plate of eggs of the frog, Xenopus laevis. Overexpression of vertebrate FoxD3 in the Xenopus neural plate results in ectopic neural crest. Thus, the vertebrate FoxD3 protein promotes the formation of neural crest at the expense of the neural plate. However, overexpression of AmphiFoxD had the opposite effect. The neural plate was expanded and expression of neural crest markers was suppressed. We conclude that the activation domain of FoxD has changed during the evolution of FoxD3 to allow a new role in promotion of neural crest. These experiments are on-going. In summary, our NASA-supported work has resulted in five papers in 2003/2004. These include a phylogenetic analysis of FoxD genes (Mazet et al., 2004), a methods paper describing techniques for expression of reporter constructs in amphioxus eggs (Holland and Yu, 2004) and a paper describing the tissue-specific expression of reporter constructs in amphioxus eggs (Yu et al., 2004). This work has shown that evolution of neural crest probably involved the evolution of new cis-regulatory elements in the FoxD3 gene after gene duplication and also involved the evolution of new transactivation properties in the FoxD protein. These results have shed light on the mechanism of evolution of neural crest and on the evolution of new structures generally.

Bibliography Type:	Description: (Last Updated: 05/04/2007)
Articles in Peer-reviewed Journals	McCauley, D.; Bronner-Fraser, M. "Expression and phylogenetic analysis suggest a single ancestral gene." Evolution and Development, in press. , Dec-2004
Articles in Peer-reviewed Journals	Cerny, R.; Lwigale, P.; Ericsson, R.; Meulemans, D.; Epperlein, H.H.; Bronner-Fraser, M. "Developmental origins and evolution of jaws: new interpretation of “maxillary” and “mandibular”. " Developmental Biology, in press. , Dec-2004
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